

1. Validation summary

Field	Value
Validation status	FAILED
Simulation readiness	Not ready
Configuration file	/home/zhangs37/ai_chainsight/workspace/fem-cs-test/scenarios/baseline-fem-hpfd-network-0386/config/baseline-fem-hpfd-network-0386.xlsx (workbook read in tolerant mode because the stylesheet XML contains invalid font-family metadata; source workbook was not modified)
Simulation period	2026-02-02 to 2026-05-03
Scope benchmark	Seed = M1_DemandForecast material-location; expanded benchmark = recursive upstream Global_Network material-location closure (seed=388, expanded=411)
Total ERROR count	38
Total WARNING count	2

2. Issue detail

Issue ID	Rule ID	Severity	Tab	Field / Scope	Key value	Issue	Expected	
ISSUE-001	SCHEMA-NUMFMT	ERROR	M1_DemandForecast	week	4959 row(s); sample index 0, 1, 2	Numeric field contains non-numeric values.	Numeric fields must be parseable numbers.	Fi nu th
ISSUE-002	SCHEMA-EMPTYCOL	ERROR	M4_MaterialLocationLineCfg	min_batch	min_batch	Required column is fully empty.	Required columns should contain maintained values.	Pr re
ISSUE-003	SCHEMA-EMPTYCOL	ERROR	M4_MaterialLocationLineCfg	rv	rv	Required column is fully empty.	Required columns should contain maintained values.	Pr re
ISSUE-004	SCHEMA-EMPTYCOL	ERROR	M4_MaterialLocationLineCfg	ptf	ptf	Required column is fully empty.	Required columns should contain maintained values.	Pr re
ISSUE-005	SCHEMA-EMPTYCOL	ERROR	M4_MaterialLocationLineCfg	day	day	Required column is fully empty.	Required columns should contain maintained values.	Pr re
ISSUE-006	SCHEMA-EMPTYCOL	ERROR	M4_MaterialLocationLineCfg	MCT	MCT	Required column is fully empty.	Required columns should contain maintained values.	Pr re
ISSUE-007	SCHEMA-EMPTYCOL	ERROR	M4_ChangeoverDefinition	cost	cost	Required column is fully empty.	Required columns should contain maintained values.	Pr re
ISSUE-008	SCHEMA-EMPTYCOL	ERROR	M4_ChangeoverDefinition	mu_loss	mu_loss	Required column is fully empty.	Required columns should contain maintained values.	Pr re
ISSUE-009	XREF-ML-MISSING	ERROR	Global_Network	material/location	36 combo(s); sample 80728859-0386, 80728861-0386, 80799571-0386	Global_Network is missing expanded benchmark material-location combinations.	Global_Network should cover the full expanded benchmark scope.	Bi be cc G
ISSUE-010	XREF-ML-EXTRA	ERROR	Global_Network	material/location	3639 combo(s); sample 80684724-A668, 80684724-A680, 80684724-A715	Global_Network contains material-location combinations outside the expanded benchmark scope.	Global_Network should align to the expanded benchmark scope.	Ri re sc lo
ISSUE-011	XREF-ML-MISSING	ERROR	M1_InitialInventory	material/location	28 combo(s); sample 80728859-0386, 80728861-0386, 80799571-0386	InitialInventory is missing expanded benchmark material-location combinations.	M1_InitialInventory should cover the full expanded benchmark scope.	Bi be cc M
ISSUE-012	XREF-ML-EXTRA	ERROR	M1_InitialInventory	material/location	36 combo(s); sample 80728859-386, 80728861-386, 80799571-386	InitialInventory contains material-location combinations outside the expanded benchmark scope.	M1_InitialInventory should align to the expanded benchmark scope.	Ri re sc
ISSUE-013	XREF-ML-MISSING	ERROR	M1_ForecastError	material/location	13 combo(s); sample 80858047-C867, 80858048-C867, 80858048-D608	ForecastError is missing expanded benchmark material-location combinations.	M1_ForecastError should cover the expanded benchmark scope defined for this validation.	Bi be cc M
ISSUE-014	XREF-ML-EXTRA	ERROR	M1_ForecastError	material/location	112 combo(s); sample 80728859-D901, 80728859-E546, 80728861-D901	ForecastError contains material-location combinations outside the expanded benchmark scope.	M1_ForecastError should align to the expanded benchmark scope defined for this validation.	Ri re sc

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ISSUE-015	XREF-ML-MISSING	ERROR	M3_SafetyStock	material/location	31 combo(s); sample 80799573-E568, 80799574-C563, 80799575-E556	SafetyStock is missing expanded benchmark material-location combinations.	M3_SafetyStock should cover the full expanded benchmark scope.	Be be cc M
ISSUE-016	XREF-ML-EXTRA	ERROR	M3_SafetyStock	material/location	2681 combo(s); sample 80684724-A668, 80684724-A672, 80684724-A673	SafetyStock contains material-location combinations outside the expanded benchmark scope.	M3_SafetyStock should align to the expanded benchmark scope.	Ri re sc
ISSUE-017	XREF-ML-MISSING	ERROR	M5_DeployConfig	material/receiving	36 combo(s); sample 80728859-0386, 80728861-0386, 80799571-0386	DeployConfig is missing expanded benchmark material-location combinations.	M5_DeployConfig receiving scope should cover the full expanded benchmark scope.	Be be cc M
ISSUE-018	XREF-ML-EXTRA	ERROR	M5_DeployConfig	material/receiving	3639 combo(s); sample 80684724-A668, 80684724-A680, 80684724-A715	DeployConfig contains material-location combinations outside the expanded benchmark scope.	M5_DeployConfig should align to the expanded benchmark scope.	Ri re sc
ISSUE-019	XREF-MATERIAL	ERROR	Global_Network	material	324 material(s); sample 80684724, 80684725, 80684726	Tab contains materials outside the expanded benchmark material set.	Maintained materials should align to the benchmark material set.	Ri re sc
ISSUE-020	XREF-MATERIAL	ERROR	M3_SafetyStock	material	285 material(s); sample 80684724, 80684725, 80684726	Tab contains materials outside the expanded benchmark material set.	Maintained materials should align to the benchmark material set.	Ri re sc
ISSUE-021	XREF-MATERIAL	ERROR	M4_MaterialLocationLineCfg	material	1 material(s); sample 90464033	Tab contains materials outside the expanded benchmark material set.	Maintained materials should align to the benchmark material set.	Ri re sc
ISSUE-022	XREF-MATERIAL	ERROR	M5_DeployConfig	material	324 material(s); sample 80684724, 80684725, 80684726	Tab contains materials outside the expanded benchmark material set.	Maintained materials should align to the benchmark material set.	Ri re sc
ISSUE-023	XREF-MATERIAL	ERROR	M6_MaterialMD	material	274 material(s); sample 80684724, 80684725, 80684726	Tab contains materials outside the expanded benchmark material set.	Maintained materials should align to the benchmark material set.	Ri re sc
ISSUE-024	XREF-LANE	ERROR	M6_TruckReleaseCon	sending/receiving	16 lane(s); sample 2799->C816, A672->C816, A672->D352	Tab contains sending-receiving combinations outside Global_Network.	Sending-receiving combinations should exist in Global_Network sourcing-location.	Ri re sc
ISSUE-025	XREF-LOCATION	ERROR	M3_SafetyStock	location	1 location(s); sample 0386	Location reference does not exist in Global_Network.	Locations should exist in Global_Network.	Ri re ne
ISSUE-026	XREF-LOCATION	ERROR	M4_MaterialLocationLineCfg	location	1 location(s); sample 386	Location reference does not exist in Global_Network.	Locations should exist in Global_Network.	Ri re ne

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ISSUE-027	XREF-LOCATION	ERROR	M4_LineCapacity	location	1 location(s); sample 386	Location reference does not exist in Global_Network.	Locations should exist in Global_Network.	Re-renew
ISSUE-028	XREF-LOCATION	ERROR	M4_ProductionReliability	location	1 location(s); sample 386	Location reference does not exist in Global_Network.	Locations should exist in Global_Network.	Re-renew
ISSUE-029	XREF-LOCATION	ERROR	M5_PushPullModel	sending	1 location(s); sample 0386	Location reference does not exist in Global_Network.	Locations should exist in Global_Network.	Re-renew
ISSUE-030	HC-GLT-001	ERROR	Global_LeadTime	OTD/PDT	52 row(s); sample C810->D767, A673->C819, 0386->A716	OTD/PDT validity rule failed.	0 <= OTD < PDT.	Fix OTD or PDT
ISSUE-031	HC-GDP-001	ERROR	Global_DemandPriority	demand_element coverage	10 missing pattern(s); sample net demand for net demand for net demand for normal, net demand for net demand for net demand for normal, net demand for net demand for net demand for ao	Demand element coverage does not span the maximum network depth.	Demand elements should cover all required demand types through the maximum network layer.	Add demand rows
ISSUE-032	HC-M1DF-001	ERROR	M1_DemandForecast	week	min=6, max=18	Forecast week coverage is incomplete for the simulation period plus 2 weeks.	week should start at 1 and end at least at 15.	Re-renew for the horizon
ISSUE-033	HC-M1OC-001	ERROR	M1_OrderCalendar	date	90 missing date(s); sample 2026-02-02, 2026-02-03, 2026-02-04	Order calendar does not cover the full simulation period.	Dates should cover every day from 2026-02-02 to 2026-05-03.	Balance the period
ISSUE-034	HC-M3SS-001	ERROR	M3_SafetyStock	material/location daily coverage	3061 combo(s); sample 80684724-A668, 80684724-A672, 80684724-A673	SafetyStock does not cover every simulation date for some material-location combinations.	Every material-location should have daily safety-stock records for the full simulation period.	Balance data rows
ISSUE-035	HC-M4MLL-002	ERROR	M4_MaterialLocationLineCfg	numeric fields	30 row(s); sample 80878361-386, 80878362-386, 80878369-386	One or more M4 numeric fields are non-numeric or blank.	prd_rate, min_batch, rv, ptf, lsk, day, and MCT must be numeric.	Fix numeric values
ISSUE-036	HC-M4LC-001	ERROR	M4_LineCapacity	location/line date coverage	1 combo(s); sample 386-HPFD	LineCapacity does not cover the full simulation period for some maintained lines.	Each location-line should cover every day from 2026-02-02 to 2026-05-03.	Balance lines for
ISSUE-037	HC-M4CO-002	ERROR	M4_ChangeoverMatrix	line pair completeness	435 missing pair(s); sample HPFD:80728859->80799571, HPFD:80728859->80799573, HPFD:80728859->80799574	ChangeoverMatrix is incomplete for at least one line.	Each line with n delegated materials should have n^2 - n non-self pairs.	Add non-ch

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ISSUE-038	HC-M5PP-001	ERROR	M5_PushPullModel	model	1 invalid value(s); sample pull	PushPullModel contains invalid values.	model can only be push or soft push.	R m p t
ISSUE-039	SC-M1INV-DFCNA	WARNING	M1_InitialInventory	DFC denominator	419 row(s); sample 80728859-386, 80728859-A668, 80728859-A672	Week 1 daily average demand is zero or missing, so DFC cannot be evaluated for part of the inventory scope.	Week 1 demand should exist when DFC is used as a soft-risk signal.	A w r o t h s t e x D
ISSUE-040	SC-M4MLL-DEMANDNA	WARNING	M4_MaterialLocationLineCfg	min_batch vs demand availability	30 row(s); sample 80878361-386, 80878362-386, 80878369-386	Near-term demand is zero or missing for maintained M4 rows, so min_batch risk cannot be fully assessed.	Week 1 to week 4 demand should exist when evaluating batch-size risk.	C i t h i n i n t o r e r o

3. Tab-level summary

Tab	Status	ERROR count	WARNING count	Main issue type	Remark
Global_Network	FAILED	3	0	XREF-ML-MISSING / XREF-ML-EXTRA / XREF-MATERIAL	Present in workbook; blocking issues were found.
Global_LeadTime	FAILED	1	0	HC-GLT-001	Present in workbook; blocking issues were found.
Global_DemandPriority	FAILED	1	0	HC-GDP-001	Present in workbook; blocking issues were found.
M1_DemandForecast	FAILED	2	0	SCHEMA-NUMFMT / HC-M1DF-001	Present in workbook; blocking issues were found.
M1_OrderCalendar	FAILED	1	0	HC-M1OC-001	Present in workbook; blocking issues were found.
M1_InitialInventory	FAILED	2	1	XREF-ML-MISSING / XREF-ML-EXTRA / SC-M1INV-DFCNA	Present in workbook; blocking issues were found.
M1_ForecastError	FAILED	2	0	XREF-ML-MISSING / XREF-ML-EXTRA	Present in workbook; blocking issues were found.
M3_SafetyStock	FAILED	5	0	XREF-ML-MISSING / XREF-ML-EXTRA / XREF-MATERIAL / XREF-LOCATION / HC-M3SS-001	Present in workbook; blocking issues were found.
M5_PushPullModel	FAILED	2	0	XREF-LOCATION / HC-M5PP-001	Present in workbook; blocking issues were found.
M5_DeployConfig	FAILED	3	0	XREF-ML-MISSING / XREF-ML-EXTRA / XREF-MATERIAL	Present in workbook; blocking issues were found.
M6_MaterialMD	FAILED	1	0	XREF-MATERIAL	Present in workbook; blocking issues were found.
M6_TruckReleaseCon	FAILED	1	0	XREF-LANE	Present in workbook; blocking issues were found.
M6_TruckTypeSpecs	PASS	0	0	None	Present in workbook and checked against the configured rules.
M6_DeliveryDelayDistribution	PASS	0	0	None	Present in workbook and checked against the configured rules.
M1_AOConfig	PASS	0	0	None	Present in workbook and checked against the configured rules.
M1_DPSCConfig	NOT MAINTAINED	0	0	Not maintained	Tab is absent or present but empty, so it was not treated as maintained for this validation pass.
M1_SupplyChoiceConfig	NOT MAINTAINED	0	0	Not maintained	Tab is absent or present but empty, so it was not treated as maintained for this validation pass.
M4_MaterialLocationLineCfg	FAILED	8	1	SCHEMA-EMPTYCOL / XREF-MATERIAL / XREF-LOCATION / HC-M4MLL-002 / SC-M4MLL-DEMANDNA	Present in workbook; blocking issues were found.
M4_LineCapacity	FAILED	2	0	XREF-LOCATION / HC-M4LC-001	Present in workbook; blocking issues were found.
M4_ProductionReliability	FAILED	1	0	XREF-LOCATION	Present in workbook; blocking issues were found.
M4_ChangeoverDefinition	FAILED	2	0	SCHEMA-EMPTYCOL	Present in workbook; blocking issues were found.
M4_ChangeoverMatrix	FAILED	1	0	HC-M4CO-002	Present in workbook; blocking issues were found.
Global_seed	PASS	0	0	None	Present in workbook and checked against the configured rules.
Global_SpaceCapacity	NOT MAINTAINED	0	0	Not maintained	Tab is absent or present but empty, so it was not treated as maintained for this validation pass.

Tab	Status	ERROR count	WARNING count	Main issue type	Remark
M6_MDQBypassRules	PASS	0	0	None	Present in workbook and checked against the configured rules.
M6_TruckCapacityPlan	NOT MAINTAINED	0	0	Not maintained	Tab is absent or present but empty, so it was not treated as maintained for this validation pass.

4. Readiness conclusion

Decision	Condition	Conclusion text
FAILED	Any ERROR exists	Configuration is not ready for simulation. Blocking issues must be fixed before running ChainSight engine. The current blockers are mainly expanded benchmark-scope mismatches, forecast week and date-coverage gaps, incomplete safety-stock / M4 coverage, and invalid lead-time or push-pull settings.
PASS WITH WARNINGS	No ERROR exists, but at least one WARNING exists	Configuration can be used for simulation, but warning risks should be reviewed.
PASS	No ERROR and no WARNING	Configuration is ready for simulation.